B.Tech Mini Project Synopsis

HTTP Security Header Analyzer

A Web-Based Security Assessment Tool for Analyzing HTTP Response Headers

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Submitted for	B.Tech CSE — 5th Semester Mini Project
Domain	Cyber Security / Web Application Security
Frontend	HTML, CSS, JavaScript
Backend	Python (Flask)
Report Generation	ReportLab (PDF)
Version Control	Git & GitHub

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1. Introduction

Web applications are increasingly becoming targets of cyberattacks such as Cross-Site Scripting (XSS), Clickjacking, MIME-type sniffing, and Man-in-the-Middle (MITM) attacks. Many of these attacks can be mitigated by configuring appropriate HTTP security headers. However, many websites either lack these headers or configure them incorrectly, leaving them vulnerable.

The HTTP Security Header Analyzer is a web-based security assessment tool that automatically analyzes HTTP response headers of a given website, identifies missing or misconfigured security headers, evaluates the overall security posture, and generates a detailed report with recommendations for improving website security.

2. Problem Statement

Many website administrators are unaware of the importance of HTTP security headers or fail to configure them correctly. Existing vulnerability scanners are often expensive or provide excessive information that may not be suitable for students or small organizations. Therefore, there is a need for a lightweight, easy-to-use tool that focuses specifically on analyzing HTTP security headers and providing practical remediation guidance.

3. Objectives

The objectives of this project are:

- To develop a web application that analyzes HTTP response headers.
- To detect missing or improperly configured security headers.
- To calculate a security score based on the analysis.
- To classify the website's security level.
- To generate a detailed security report with recommendations.
- To improve awareness of web security best practices.

4. Scope of the Project

The proposed system will:

- Accept a website URL as input.
- Send an HTTP request to the target website.
- Retrieve HTTP response headers.
- Analyze important security headers such as Content-Security-Policy (CSP), Strict-Transport-Security (HSTS), X-Frame-Options, X-Content-Type-Options, Referrer-Policy, and Permissions-Policy.
- Detect missing or insecure header values.
- Generate a security score and risk assessment.
- Produce a downloadable security report.

5. Proposed Methodology

The system follows these steps:

- User enters the website URL.
- Backend sends an HTTP request.
- HTTP response headers are collected.
- Security headers are analyzed.
- Missing or weak configurations are identified.
- Risk level is calculated.
- Security score is generated.
- Final report with recommendations is displayed.

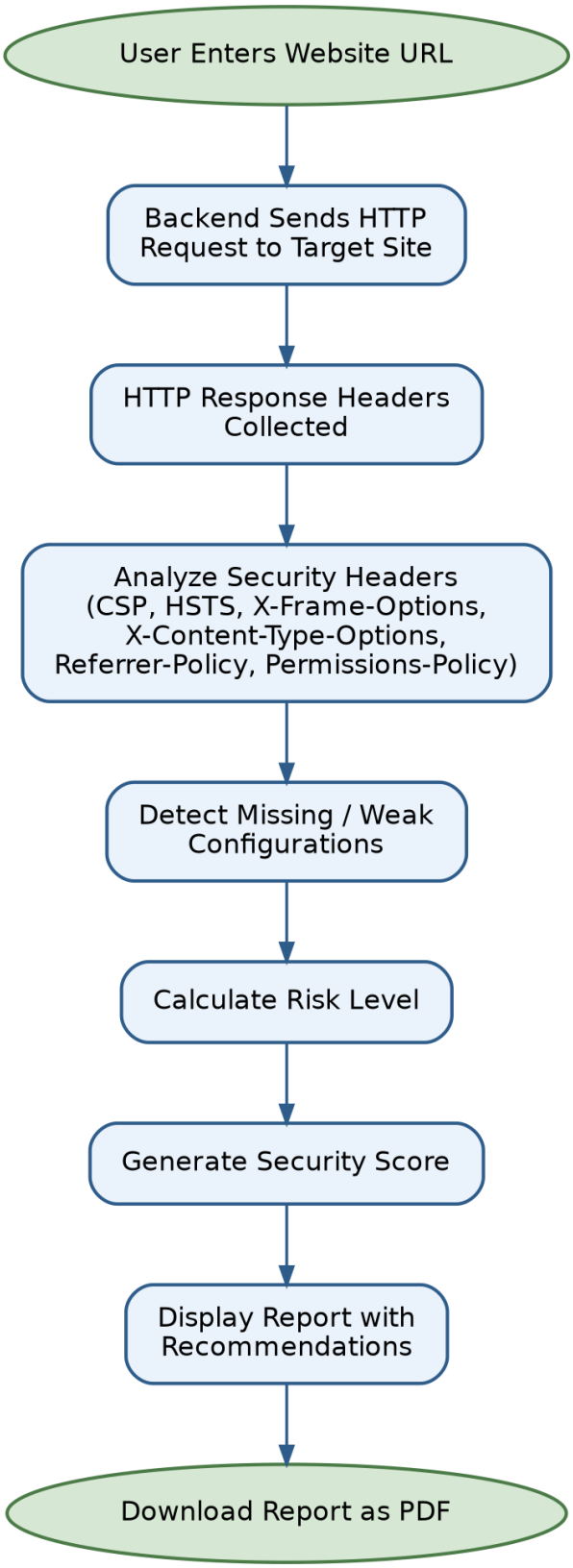


Fig. 1 — System Workflow / Flowchart of the HTTP Security Header Analyzer

6. Functional Requirements

- Website URL input
- HTTP response header retrieval
- Security header analysis
- Misconfiguration detection
- Security score generation
- Risk classification
- Report generation
- Download report as PDF

7. Non-Functional Requirements

- User-friendly interface
- Fast response time
- High reliability
- Secure processing
- Cross-platform compatibility
- Easy maintenance

8. Tools and Technologies

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python Flask
Language	Python
HTTP Library	Requests
Report Generation	ReportLab
IDE	Visual Studio Code
Version Control	Git & GitHub

9. Expected Outcomes

The system will:

- Detect missing HTTP security headers.
- Identify weak security configurations.
- Generate a comprehensive security assessment report.
- Provide recommendations for improving web application security.
- Help developers follow secure web development practices.

10. Applications

- Website security assessment
- Educational cybersecurity laboratories
- Web development testing
- Security auditing
- Learning HTTP security best practices

11. Future Enhancements

- OWASP Top 10 vulnerability checks
- SSL/TLS certificate analysis
- Cookie security analysis
- Multiple website scanning
- Scan history dashboard
- User authentication
- REST API integration
- Automated scheduled security scans

12. Conclusion

The HTTP Security Header Analyzer is a lightweight web security assessment tool designed to evaluate the security configuration of websites through HTTP response headers. It helps developers and administrators identify missing or misconfigured security headers and provides practical recommendations to improve website security. The project promotes secure coding practices and contributes to reducing common web application vulnerabilities.

References

1 Mozilla Developer Network (MDN) – HTTP Security Headers
2 OWASP Secure Headers Project
3 OWASP Top 10 Web Application Security Risks
4 RFC 6797 – HTTP Strict Transport Security (HSTS)
5 Python Requests Documentation
6 Flask Documentation